

A Conceptual Framework for Managing Service Desks: A South African Perspective

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Abstract--The service desk is a critical part of any organization, as it is in most cases the first point of contact with customers. Therefore, it holds the key to the organization's profitability and survival. A qualitative interpretive research approach was followed in order to get a better understanding of the real problems staff and users of such desks experience. Field studies as well as short case studies are used to analyze the typical service desks in the South African business environment with special emphasis on the role of information technology. Based on previous research in the field of relationship management, the article presents a conceptual framework to improve the current situation. The framework explains the elements to be focused on to ensure that staff manning such desks, are effective in their operations. The framework also serves as a "tool" for managers to support them in their decision making regarding the operations of such desks.

I. INTRODUCTION AND BACKGROUND

Service desks involve the use of proper technology by organizations to present a contact point for services being rendered to customers. The role and impact of Information and Communications Technology (ICT) and Information Systems (IS) in today's organization cannot be over-emphasized. In this regard Corea [10] notes that:

"Information technology systems may be considered to a certain extent as pre-formulated designs for organizational action, as they are used by firms to deliberately structure or prescribe the way work is done, or the way interactions are carried out with customers, suppliers and other parties."

Various research studies have endorsed the use of ICT systems for gaining competitive advantage. Heras *et al* [17] note that differentiation between competing companies is difficult nowadays, as products, prices and quality are very similar. They explain that offering careful attention to their customers is an important strategy used by companies to try and obtain an advantage over their competitors. In a digital economy where users directly access e-services such as e-commerce, online banking, etc, the possible impact of IS to economic development cannot be over-emphasized. However, with the continuing growth of e-services the need for "service desks" is increasing in order to assist both contact and e-customers. This has led to dramatic increase and use of "service desks" and "call centers".

Call centers are described by Corea [10] as "a modern IT based organizational form that is at the forefront of a current trend of technological intensification of service activity". However, it is also noted by Corea [10] that despite their

wide, dramatic rates of adoption call centers have had limited attention in recent IS research.

II. PROBLEM DESCRIPTION

Leonard [29] points out that, since the advent of the Internet, Internet-based Self Service Technologies (SSTs) have been adopted widely in the private sector. A number of studies in the private sector environment have already demonstrated that, despite their growth, the acceptance of these SSTs by end users has been mediocre [6]; [7]; [33]. According to Naidoo [29] these findings are congruent with findings in some of the healthcare initiatives, who have acted on the potential opportunities presented by these service innovations [2]. So far most research that has examined the slow acceptance rate, has followed a reductionist approach – concentrating either on the individual user, the technology or the organisational context. More specifically, little is known about how self-service technology implementations are shaped by social, political, economic and technological issues.

The South African business environment is also not left out in the quest for improved services through the use of call centers and service or help desks in both "contact service" organizations and "electronic service" organizations.

Naidoo [29] also mentions that a large number of firms moving to Internet-enabled customer service should also be aware of several myths. The first myth proposes that the Internet can minimise the cost of providing customer service [13]. In contrast, recent evidence suggests that it is generally more costly to design and maintain an effective electronic customer service [33]; [24]. The argument is that Internet customers have higher expectations and are more informed, and as a result they expect to be serviced by more knowledgeable and expensive call-centre representatives. Therefore, instead of replacing traditional channels, Internet-based services tend to be encouraging more advanced human support. Furthermore, the data infrastructure required to support a comprehensive online self service strategy is often more complex than initially envisaged and requires high investment and cooperation at industry and intra-organisational level [23].

Another myth (according to Naidoo [29]) is the suggestion that Internet-based services will facilitate the process of customer relationship management [35]; [5]. Recent research suggests that potentially valuable customers may perceive barriers to interaction with technology-enabled service systems [38]. Advocates of this view argue that the electronic service delivery process often does not address the

various needs, capabilities and concerns of the user, as they are designed mainly with the aim of achieving operating efficiencies for the organisation. Furthermore, organisations appear to be paying little attention to existing relationships which the technology wishes to replace [3]; [38]. In addition, a number of investigations reveal that customers are unwilling to replace face-to-face contact with electronic alternatives. It is not surprising that another stream of research indicates that face-to-face relationships may be more cost effective than virtual relationships [16].

Recent studies show that rational and economic models of electronic help desk environments are an oversimplification of what actually happens in the social environment [22]. It appears that organisations tend to ignore important contextual dynamics which may provide a deeper understanding of self-service technology implementation. Therefore this article proposes that an alternative way of conceptualising the problem is to view Internet-enabled services as a social, economic, political and technological phenomenon.

The aim of this article is to give insight into the basic problems experienced with service desks in the South African environment and to unveil their most critical shortcomings which normally directly impact on the competitive advantage of the organisation. In doing so, a conceptual framework will be proposed based on the work of Leonard [27] which will support the management service desks.

The research question that has to be answered is: What should the nature be of a typical service desk in the South African business environment in order to provide a satisfactory quality service to customers?

III. THE RESEARCH APPROACH

A qualitative research approach was followed with the philosophical assumptions of that of interpretivism. Ngwenyama *et al* [31] note that interpretivism focuses on the development of sound explanations and understandings of research phenomenon of interest. Myers [28] states that interpretive researchers tend to focus on meaning and context. According to him “they aim to understand the context of a phenomenon, since the context is what defines the situation and makes it what it is.”

Interpretivism contends that “either reality itself is a social construct or that at least our knowledge of reality is socially constructed or gained through social constructions” [36]. Stahl [36] further explains that interpretivism emphasizes context and singular occurrences in the hope of extracting meaning and making sense using qualitative methods by leaning on phenomenology (a first-hand description of human experience).

However, Leedy *et al* [26] indicate that qualitative research assumes that the researcher’s ability to interpret and make sense of what is observed is critical for understanding any social phenomenon. Therefore, the research orientation of this topic assumes that the reality for different situations could differ.

Apart from the typical literature study, the research methods that were applied are case study research as well as some field studies. Case studies are used to emphasize the typical issues with regard to poor services during service encounters. Case studies also support the research in creating a better understanding of how service desks in the South African business environment impact on competitive advantage and customer satisfaction.

Structure of Paper

In the rest of the paper a literature review is given, detailing the theoretical perspectives on quality service, expectation management as well as previous efforts towards improving customer support. Thereafter the paper describes different real life cases to illustrate the problems usually experienced by users and businesses with regards to service desks. In the following section the conceptual framework is proposed based on the work of Leonard [27], after which a concluding summary of the paper is given.

IV. LITERATURE REVIEW

A. Theoretical perspective on quality service

As noted in the introduction, the availability of many competitive services makes it easy for customers to seek better services elsewhere if they sense lack of quality service and support. Therefore, various research studies have been conducted on quality services in organizations. Reference [4] note that while one of the primary reasons for organizations to invest in Information Systems (IS) is the improvement of the quality of services (which is an important output of IS), it is noteworthy that improvement in IS has a positive correlation to the quality of service. Aside from this direct impact, it is also noted that quality IS will lead to quality information which is of strategic importance in running organizations. Moreover, the ease with which the system can be used by customer service representatives helps them to give quality service by servicing customers better with the possibility of enhancing profit.

Grandey *et al* [15] note that another key dimension of service interaction quality is the service providers’ positive attitude and demeanor, as positive displays and friendliness enhance overall service quality. Yet, Stuart *et al* [37] note that service organizations spend too little effort and resources on planning for service quality, leading to a “cycle of service failures” and lower profitability. Serious minded and forward looking organizations obviously need to avoid failures and lower profitability by taking service quality very serious. Heskett *et al* [18] demonstrate such seriousness about quality service by stating that high quality support services and policies that enable employees to deliver results to customers lead to satisfied, loyal and productive employees; employee satisfaction creates value in services provided to customers; value leads to customer satisfaction; customer satisfaction leads to loyalty; and loyalty by customers stimulates growth and profit in the organization. It is on this note that it

becomes critical to understand every step of the way that will lead to customers' satisfaction. This in essence implies understanding what the customers expect and how to manage and meet those customers' expectation. This is addressed in the next section.

B. Theoretical perspective on customers' expectations and the management thereof

Expectation is described by Coye [12] as "what customers predict about the occurrence of the service (will happen) and what customers believe about the capability of the provider (should happen)". In defining expectation in terms of beliefs, the definition of customer expectations is further refined as [12]:

"a belief about a future event that is based on information gathered directly by the customer through personal observation, indirectly through attention to information provided by others, or is inferred based on information or observation of some related occurrence".

Heskett *et al* [8] note that it is very costly to obtain new customers and that the profitability of a loyal customer grows with the time of the relationship. Therefore it is important to continually manage the customer's expectations to cultivate longstanding loyalty and retention as a key to profitability.

Towards a discussion on how customers' expectation should be managed, Coye[12] highlights the "gap" model of service quality as proposed by Zeithaml *et al* [40], which focuses on the difference between the creation of expectations and the ability of the delivery system to meet those expectations. This therefore, as inferred by Coye [12], emphasizes the existence of a number of interfaces in the overall service delivery process wherein the degree of congruence between facets will affect the final outcome. Using the example of "the relationship between descriptions in promotional materials and the actual capability of the delivery system", Coye [12] establishes the existence of expectation as the customer enters the delivery system, and the cues which may influence his/her assessment of the ongoing process of service.

To effectively manage customers' expectation, Coye [12] concludes that:

"It is incumbent on managers of service operations to be aware of the factors that influence both customer desires and customer expectations and the potential opportunities for interventions that could lead to more effective management of these relationships..."

If we have reason to believe, for example, that initial expectations are unrealistically high it may be prudent to attempt to modify these expectations to more closely fit what we know can be delivered."

Interventions may therefore include such actions as clarification, explanation, or even denial as well as transparency such as sometimes encouraging customers to

increase their level of participation. Therefore, Coye [12] highlights the importance of communication and interpersonal skills in managing expectation.

The issue of managing customer expectations could therefore be regarded as one of the corner stone management activities to maintain and improve services at help desks.

C. Previous efforts towards improving customer support

Heras *et al* [17] describe the use of Case-Based Reasoning (CBR) systems in Technology Management Systems (TMS) to control and improve processes involved in the provision of technological and customer support services. The CBR is aimed at storing and later reusing the solution applied to specific problems and the information about the problem-solving process towards improving the service or customer support offered. The ultimate goal is to shorten the solving time as Heras [17] explain that "a quick and accurate response to the customers' problems ensures their satisfaction and a good reputation for the company" which can therefore increase profits.

Ngai *et al* [30] stress the importance of data mining techniques to customer relationship management (CRM) systems. This is found important because many organizations already have a wealth of stored data about their current and potential customers and other stakeholders such as suppliers and business partners. Analyzing such database to extract useful information and subsequently gain knowledge with the use of appropriate tools and techniques in CRM is noted by Ngai *et al* [30] as worth pursuing in a customer-centric economy.

Corea [10] proposes the use of a specific database with CRM system in service desks for "such actions as recording notes on calls or searching for details of previous customer interactions". This would allow a group of customer service staff to track their efforts addressing a complaint by a customer and to coordinate and progress a situation towards its resolution over time. Another such system proposed by Corea [10] involves the use of a Lotus Notes system for holding procedure guidelines and policy documents utilised in relation to queries handled by the service desk.

V. PROBLEMS EXPERIENCED WITH REGARD TO SERVICE DESKS

In the first part of this section short cases are presented to illustrate the typical problems customers experience at service desks. In the second part specific behavioural issues at service desks are addressed.

Government Utility Case

A filing system for capturing tax records was recently introduced by a government department, obviating the need for the previous manual system. A "service-desk" was also instituted to support the users of the system.

After having logged in successfully onto the system, many users experienced trouble doing certain transactions.

Having the “peace of mind” that the helpdesk is there to assist, many people were very disappointed, because the phone was giving an “engaged tone” most of the time. Although some sent e-mails to the “help desk” out of desperation, they didn’t receive any feedback. Furthermore, penalties were given for late submissions which add frustration and impacted negatively on all customers.

The launch of such an important system needs a strong backup in terms of a very effective service desk, otherwise the service will not be used, and will be viewed as useless.

The Airline Poster Case

An airline displayed a poster indicating that users should not turn their lack of planning into emergency on the part of the service provider. An investigation into the expectation of some users reveals that most users look up to the service provider to solve their problems, regardless of whether the problems were caused by lack of planning on their side, or by other reasons. More specifically, most users are willing to pay a little extra to have their problem solved rather than to take certain precautions such as “booking flight 2 weeks in advance”. These users believe life to be full of various intricacies that make it difficult to adequately plan ahead and stick to such plans.

Interestingly, on a follow up trip to the same airline company to obtain management’s view of such poster, the author came across another poster. This time, the poster is not merely typed, but color printed with logos of the company, indicating management’s approval of such poster. The poster states that:

“We understand that traveling can be a stressful experience but please don’t take it out on our staff”

When some travelers were questioned on this poster, the general consensus is that they expect airline companies in general to say something like:

“We understand that traveling can be a stressful experience but rest assured that our staff will do their utmost to relieve you of the stressful experience. Kindly help them to assist you.”

The Airline Service Encounter Case, based on the work of Woodcock [39]:

Although this case is not part of the “South African” environment, it illustrates a typical bad service encounter “story” that could have happened anywhere in the world, also in South Africa.

“We began writing this chapter on a flight from London to New Zealand, via Denver. The journey had started badly. Mike had driven a hundred and fifty miles down congested roads. We arrived just in time for the flight.

As we struggled onto the plane the cabin staff watched us with laconic disinterest. Dave was carrying his portable computer, a large bag of papers and a suit bag. But we were looking forward to the luxury of faultless first-class airline service, imagining that our journey would be like a TV

advertisement. Needless to say, the opposite was the case. We were shown to our seats by a flight attendant who behaved with the warmth of a prison warder. She used the ‘correct’ phrases, no doubt taught to her in airline school, but the demeanor was jaded and her concern was plastic. We began to get settled for the twenty-five-hour flight. Dave’s bag would not fit into the front wardrobe, and the flight attendant barked ‘Take it to the larger closet by the exit’. Then she vanished, leaving him to struggle with an overstuffed garment bag against the human tide. Meanwhile Mike’s every request to the cabin staff met with grudging acquiescence. During the journey we noticed several minor incidents, innuendoes, oversights and slipshod errors of service. We both felt like passengers on a commuter bus journey, not first-class international jetsetters. From one point of view the blemishes were minor, and it would have seemed petty to complain, but our memory of the flight is sour and it is doubtful whether we will willingly fly this airline again. In a hundred ways the flight staff ‘did the wrong thing’.”

This case emphasizes Cortada’s [11] comments on the importance of the moment when a service is performed:

“... since consumption of a service occurs as it is performed, consumption and production are inseparable and perishable - they occur once, as an airplane flight or a restaurant meal. Customer-oriented definitions therefore call for focus on the reliability of services provided, willingness or responsiveness to service someone quickly, ability of employees to offer trust and confidence, empathy, and a good business environment.”

Behavioural issues reported by customers with regard to service desks

Apart from response delays in attending to requests, customers usually express frustrations in even logging their requests in the first instance. Various customers indicate that one of the reasons they don’t like communicating with service desks, is that one is usually directed to talk to machines or recorded messages which ask customers to press certain keys for certain options, and after that are presented with another set of keys for another set of options, etc. When such calls are made during the day, it costs the customers a lot. The worst part is that sometimes, before concluding with the service desks, the telephone line may drop, maybe due to insufficient fund, and when the call is remade, one has to start afresh.

The following is a summary of the most important issues reported by customers and businesses with regard to the use of service desks:

- Inability of service desk staff to provide a *one-stop service*.
- Customers sometimes experience *inauthentic display* during service encounters. In this regard it is important to take note that research has demonstrated the ability to distinguish authentic smiles from inauthentic smiles and

as a result infers that customers' reactions to inauthentic displays are less positive than to an authentic smile [15].

- *Emotional customers.* In such cases the service providers are expected to do everything possible to please the customers because as noted by Grandey *et al* [15], "once an unhappy or dissatisfied customer walks out the door, they are gone forever!" Apart from this these customers normally do further damage to the company by spreading the "bad experience message".
- *Culture differences.* Companies need to be very sensitive when dealing with a large variety of cultures. Grandey [15] note for example that a service provider who makes eye contact is likely to be perceived as honest and friendly. However, from the Central and West African culture, an actor that makes eye contact with elderly people is considered disrespectful. Much more respect is demanded if the elder is a client. In this regard Grandey [15] state "In service encounters, the purpose of the encounter is to receive some product or service; thus, many other factors beside interpersonal demeanor are also critical to target reactions."
- *Nonchalant attitude and incompetence* of service desks' staff. Often, either due to fatigue, incompetence or other forms of lack of motivations, service desk staff behave in a way that put customers off. Inadequate funds often force businesses to employ less qualified staff which are paid a smaller salary and end up being non-motivated. Unfortunately "there may be only one opportunity to win over a customer in a service encounter", and if that one opportunity is missed, customers may conclude negatively about the whole organization without knowing there could be other "good" employees at the back offices in the organization [15].

VI. CONCEPTUAL FRAMEWORK FOR MANAGING SERVICE DESKS

The proposed conceptual framework is adapted from the work of Leonard [27]. He proposes a conceptual framework for the establishment and maintenance of sound relationships between IT professionals and end users. The basic framework is illustrated in figure 1. A key element of service desks or service encounters is to establish sound

relationships. It is argued that this framework can be adapted in such a way that it can serve as a convenient way of explaining what components of activities are paramount for any service desk to act as "front line" for any organization.

In the following paragraphs the service desk operations are discussed using the framework of Leonard [27] and thereafter an adapted framework is proposed showing all the components needed to operate and manage a typical service desk.

The proposed conceptual framework consists of the following three main components, each of which has sub-components explaining the internal operations of each process:

- initiation process
- activities process
- Feedback and control process.

The components in figure 1 present the life cycle of the Service Desk/Encounter. The activities involved in each component are intended to enable service staff to provide service of a high quality. In the rest of the paper the purpose and activity nature for every component is described briefly.

The summarized conceptual framework is given in the following figure.

A. Initiation process

The initiation process takes place when a service staff member gets involved in the "front line" of service. The parties (read management and staff) start with transacting, whereby information is used and produced to determine the basis of the needs of the staff members who should man the service desk. To be more specific, the information is used for the following purposes: it is a means of searching for the right people; it signals willingness to exchange; it contains the terms of the contract; it is the medium of control, enforcement and modification of the contract; and it maintains communication during bargaining [9].

The information used and produced in this process of transacting forms the building blocks of what is termed common grounds. This information plays an important role during the whole process of getting mutual understanding between management and staff members.

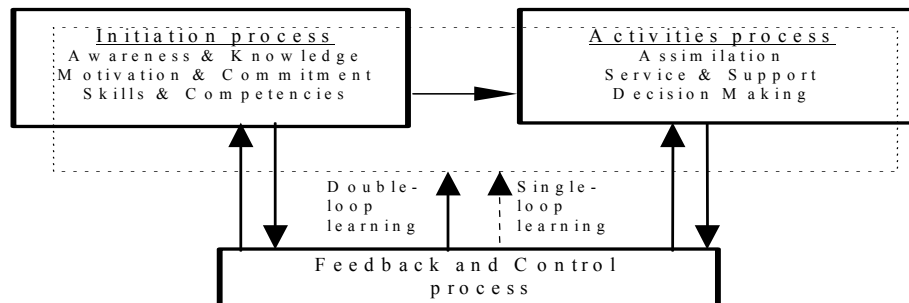


Figure 1: Conceptual framework for managing relationships between end users and IT professionals [27]

As such, a service desk of high quality involves a cultural transformation that includes changes in:

- Awareness and knowledge. Achieving an awakening and understanding of the concepts of the new paradigm.
- Motivation and commitment. Developing a new shared vision, orientation, and confidence to provide service of a high quality.
- Skills and competencies. Building the expertise necessary for creating trust and confidence.

Training and experience form the foundation for the successful implementation of a service desk and is essential to focus on when the initiation process starts.

B. Activities process

During this process three major activities take place which have an impact on the quality of the service being offered to customers:

- assimilation of service desk environment and co-workers
- service and support
- decision making process.

Although the participants may know one another by this time, they have not worked in a team yet. The purpose of the assimilation process is to allow all team members to go through an orientation process in order to understand the goals, expectations, mission, norms and ground rules (ethical issues) that might be applicable when performing duties at the service desk. Kelly [21] calls this process the stages of team development.

The service and support activity relates to all activities required to ensure that the customer is satisfied with what he/she has received during the service encounter. One of the biggest challenges during this process is to create trust.

The other very important activity is that of the decision making process. Apart from its normal day to day operational importance, the decision making process plays an important role in the service encounter and as such has a direct impact on the soundness of relationship building. In this regard the important role of the “lens effect” has to be addressed.

According to Pitt *et al* [32] the “law of the lens” effect will be at its strongest in dyads. They state that Brunswik’s theory of probabilistic functionalism shows us that much decision-making behaviour is best conceptualized by understanding the relationships between two participants (called systems by Pitt *et al* [32]). One system is the so-called ‘real’ network of relations between cues in the environment and the event to be predicted. The second system is the network between cues in the individual’s mind and his or her predictions. This stresses the importance of competent staff manning service desks.

It is noted by Grandey *et al* [15] that impression management should be a critical part of the service provider’s role in service encounters, as service encounters personnel

are often the first and only person with whom the public ever interacts. The company should therefore create a vested interest in impressions being well managed. This could explain why some organizations enforce display requirements such as appointing greeters, providing smiles and eye contact, etc. Grandey *et al* [15] further note that employees should go further than the minimum requirement of “simply pasting on a fake smile” but should put in an effort to achieve an authentic positive display by remaining positive and enthusiastic throughout the workday so as to increase the overall value of the service encounter.

Personal touch is found by Babbar *et al* [1] to substantively affect customer satisfaction. Therefore, individual attention, helpfulness, courtesy and promptness are strongly recommended in enhancing service.

According to Grandey *et al* [15], impression management theorists have confirmed that “one can strategically create a positive impression by appearing both competent and likeable”. Expertise or competence is therefore noted as one of the dimensions of interaction quality of service encounters. Therefore employees can “manage impressions of competence by being efficient, available, accurate, and knowledgeable about products”.

C. Feedback and control process

This is a continuous process which gives all participants the opportunity to give their own evaluation (feedback) of how they experience the activities of the service desk. “Individuals learn as a result of their experience, and so do organizations” [34]. Furthermore, it allows participants to have control over the progress of a specific service and support activity. This process gives all participants involved the opportunity to give feedback on their experience at the service desk. Suggestions on how a specific process of the conceptual framework (cf figure 1) in general could be improved should also form part of this ongoing process.

Viewing the feedback process as a process of learning, two well-known types of learning are relevant, namely single-loop learning and double-loop learning. There is a wide set of ideas linked to these two types of learning. Robey *et al* [34] explain it by making use of the term organizational memory, which they define as: “frames of meaning that are shared by an organization’s members.”:

“...[R]evisions of memory produce what has been called ‘double-loop learning’, or ‘higher-level learning’. In contrast with single-loop or lower-level learning, which focuses on the repetition of past behaviors, higher-level learning involves introspection into the methods by which an organization learns. If revisions of memory are guided by causal reasoning that incorporates complex rules and associations regarding new actions, then the organization can make sense of its past experience. Such learning requires an understanding of the reasons for following a new course of action, not just the routines involved in that action.”

Introna [20] states that in the case of single-loop learning the theory-in-use is not adjusted. The actor merely adjusts his or her action without attending to the theory-in-use. The theory-in-use is implicit and invisible. In general terms, the structure of the system remains unchanged over time and becomes rigid, although there may be some localized changes. In the case of service desks it means that the participants may for example change their behaviour, way of communication, co-operation approach, etc., in order to improve the quality of service from their side.

In the case of double-loop learning, the theory-in-use is continually adjusted. The structure of the system is dynamic in local and general terms. The system “maintains” itself over time and space. This means for example that the way in which service desks are defined should be changed in order to accommodate a new paradigm of service and support.

This process therefore provides for single-loop learning and double-loop learning both during the initiation process and the activities process. In terms of single-loop learning, measurement of the attainment of objectives are used to adjust activities, whilst in the case of double-loop learning assumptions, norms and standards are continuously revised and adjusted, if necessary.

Furthermore, whenever new products and services are introduced by an organization because of competition or customer power, staff members of the service desk have to be aware of these changes and where necessary be (re)trained to cope with the demand of such new products or services. In this regard Porter’s model of five forces that impact on

organizations are applicable [25]. Businesses normally operate in rapidly changing environments which necessitates continuous change and adaption.

Based on the above mentioned reasoning and issues involved in the typical service desk environment, the following adapted conceptual framework is proposed for the operations and management of service desks (figure 2).

VII. CONCLUSIONS AND RECOMMENDATIONS

Understanding those elements that are crucial to the implementation of an effective service desk system is difficult to assess in isolation, since they represent complex interactions that can only be understood over time. The different case studies and general descriptions of problems experienced with service desks are typical of situations which the average person can identify himself with.

It is therefore recommended that organizations acknowledge and address the critical nature of the service desks in the daily activities of the organization. This implies addressing the problems faced by customers and by the business itself as was described in the different short case studies and problems experienced with service desks environments. The principal goal of service desks should be to strive to meet and if possible, exceed customers’ expectation. Coye [12] explains the need for interventions at critical points aimed at managing expectation by reducing the potential gap between customer expectations and perceived service delivery.

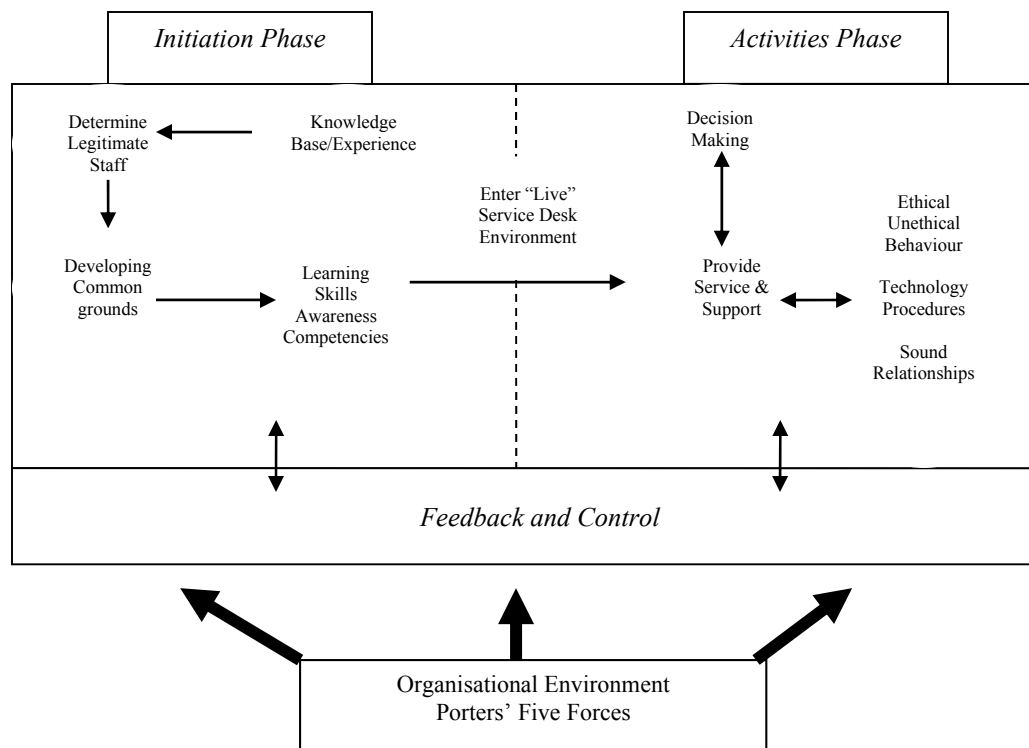


Figure 2: Conceptual framework for the operations and management of service desks (Based on the work of Leonard [27])

It is argued that if the principles of the proposed conceptual frame work are applied in the correct way and management pays attention to its different components and internal factors, service quality and the meeting of client expectations should prevail. It may be more appropriate to conclude with some parting advice for proponents of consumer-driven service desks in the eloquent words of Giddens [14]:

“No matter how well a system (read service desk) is designed and no matter how efficient its operators, the consequences of its introduction and functioning, in the contexts of other systems and of human activity in general, cannot be wholly predicted. One reason for this is the complexity of systems and actions that make up world society ... Even though we ourselves produce and reproduce it in our own actions, we cannot control social life completely” [14].

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